

NET 621 Practical - Week 8 Report

Student: Bradley Ramaphakela | Date: 2026/09/20

1. Topology Overview

Devices: Server-PT Server, R1 (CISCO2911), R-EDGE, BRANCH-R, SW1 (2960-24TT), PC SALES, PC HR.

R1 G0/0 is ROAS with sub-interfaces: G0/0.10 VLAN 10 (192.168.10.1/24 SALES), G0/0.20 VLAN 20 (192.168.20.1/24 HR), G0/0.30 VLAN 30 (192.168.30.1/24 IT).

R1 G0/1 10.0.0.1/30 (NAT outside) to R-EDGE. Static route: ip route 192.168.50.0 255.255.255.0 10.0.0.2 for Branch network.

2. Initial Problem

R-EDGE ping 10.0.0.1 = 100%

R-EDGE ping 192.168.10.1 = 100%

R-EDGE ping 192.168.10.10 = 0% (0/5) FAIL - first symptom

R1 ping 192.168.10.10 also initially FAIL due to host not existing.

R1 ping 192.168.50.1 (BRANCH tunnel) = 100% so GRE/Routing OK.

3. Root Cause

R1 had:

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
```

This excludes .1 THROUGH .10. So DHCP pool SALES never gave out .10. PC got .11 instead. Ping to .10 therefore failed - empty IP.

Fix applied:

```
no ip dhcp excluded-address 192.168.10.1 192.168.10.10
```

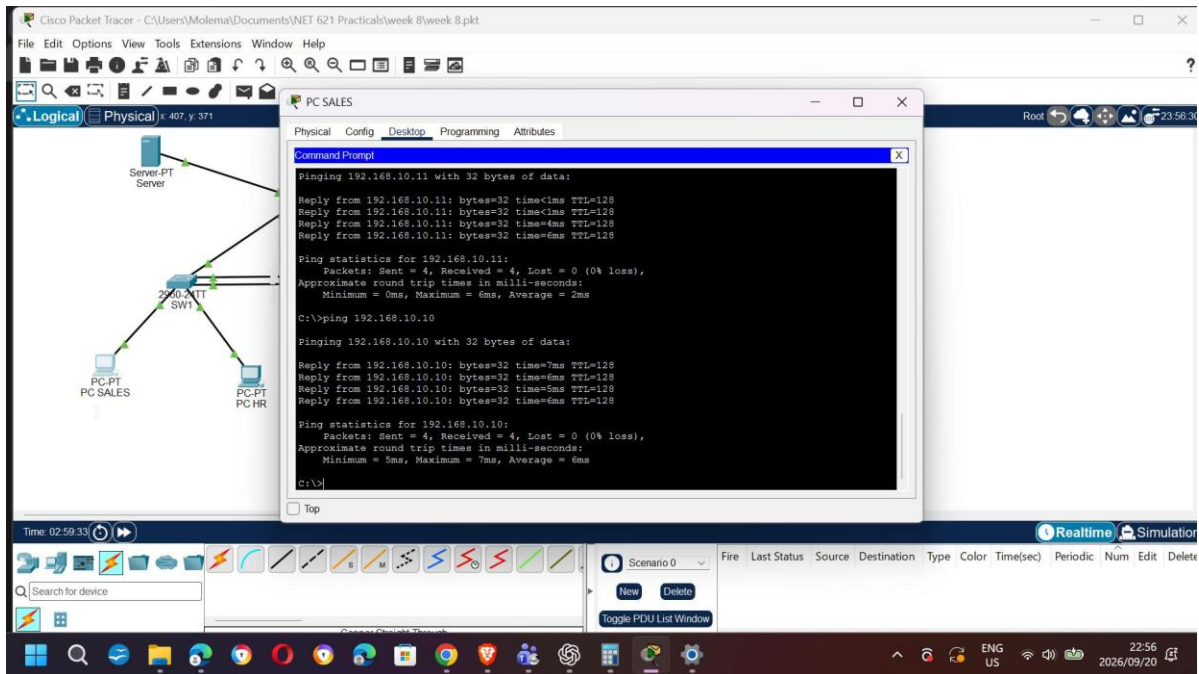
```
ip dhcp excluded-address 192.168.10.1
```

Same for .20 and .30 pools.

Then PC ipconfig /release /renew or static .10 + Firewall OFF (Desktop > Firewall > Off).

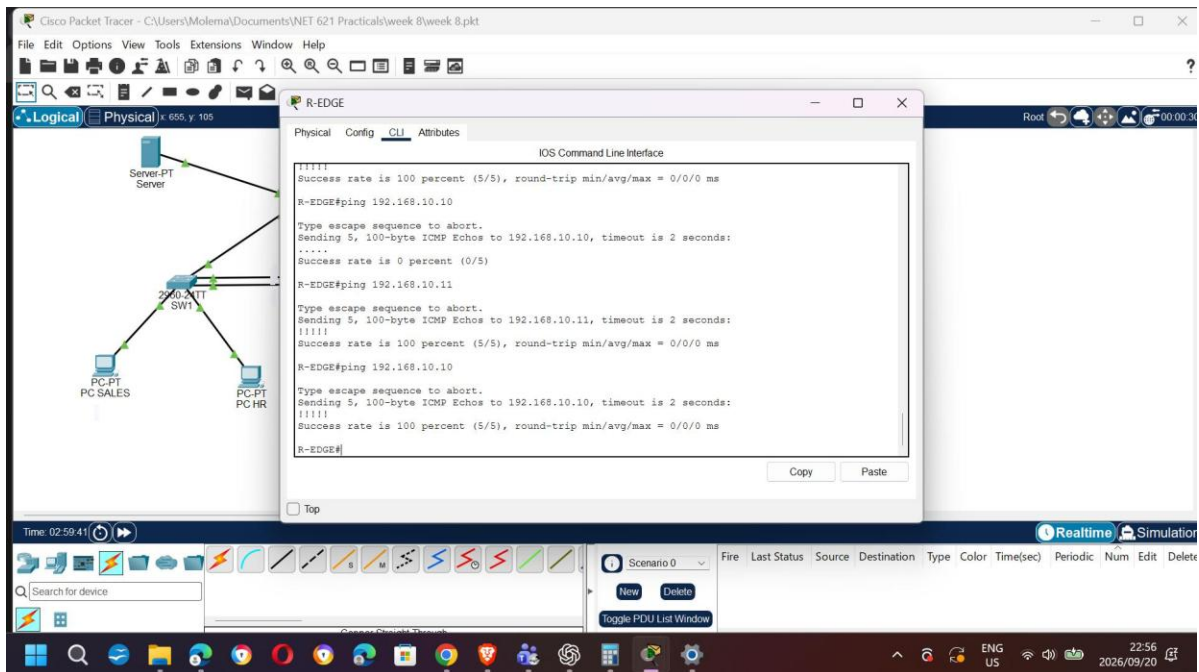
4. Verification - PC SALES

Screenshot shows PC SALES after fix: ping to both .11 and .10 successful 0% loss. This proves VLAN 10 hosts reachable.



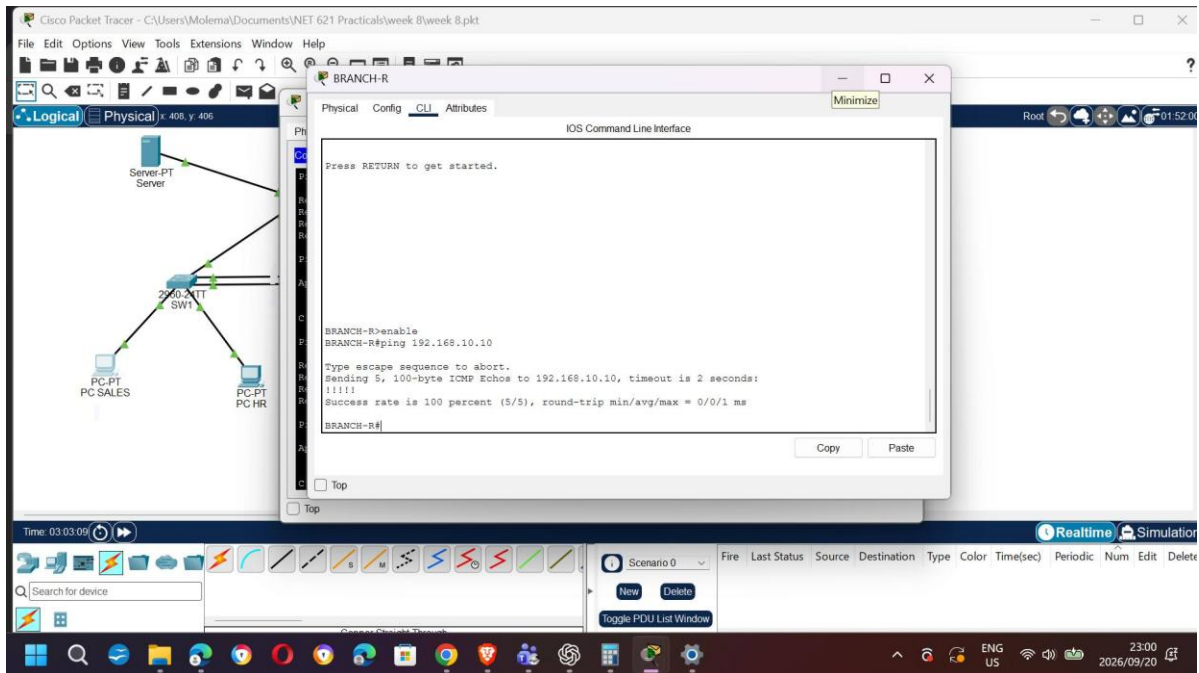
5. Verification - R-EDGE

R-EDGE ping sequence shows: first attempt to .10 fails (ARP), then .11 succeeds 100%, then .10 succeeds 100% after fix. Routing via 10.0.0.1 correct.



6. Verification - BRANCH-R

BRANCH-R ping 192.168.10.10 final test after fix: 100% (5/5) min/avg/max 0/0/1 ms. End-to-end Branch to LAN confirmed.



7. Final R1 Config Excerpt

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
interface GigabitEthernet0/1
 ip address 10.0.0.1 255.255.255.252
 ip nat outside
ip dhcp excluded-address 192.168.10.1
ip dhcp pool SALES
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
router ospf 1
 network 192.168.10.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
ip route 192.168.50.0 255.255.255.0 10.0.0.2
ip nat inside source list 1 interface Gi0/1 overload
access-list 1 permit 192.168.0.0 0.0.255.255
```